

CS7.303: Digital Signal Analysis

End Exam, Spring 2026

Time: **90 minutes**

Total Marks: **75**

- Answer all the questions, Marks for each question is provided in the brackets.
- Calculators are allowed.
- Clearly state the assumptions if any that were considered to solve.

1. What are the characteristics of a linear phase filter and describe its significance precisely with an example or necessary derivations?

[5 Marks]

1 mark → **Characteristics:** A linear phase filter has a phase response that is a linear function of frequency. This implies that the delay is constant across all frequencies.

Significance:

- Constant Delay: All frequency components of the input signal undergo the same amount of time delay. [Need to show this with an example or derivations] 2 marks
- Waveform Preservation: Because there is no phase distortion, the shape of the signal in the time domain is preserved. [Can be argued with the previous example] 2 marks

2. Can the convolution be implemented for any arbitrary impulse response ($h[n]$) of infinite length? Then, how an infinite length of $h[n]$ can be realizable in digital systems? Is there any constraint on $h[n]$ that can make the realization possible?

[5 Marks]

1 mark → **Arbitrary Convolution:** Convolution cannot be directly implemented for an arbitrary infinite because it would require infinite memory and computations per sample.

Realization in Digital Systems:

2 marks → Infinite length is realized using **Recursive Systems** (Difference Equations). Instead of a direct summation of the impulse response. [Any recursive form is fine]

Constraints:

2 marks → The impulse response has an exponentially decay properties is one such constraint. [If it is shown to lead a recursive equation, then it is good]

3. Show the DTFT characteristics of an ideal band-pass filter. Can this filter be physically realizable with digital systems? If yes, please justify the answer else suggest methods to overcome this.

[5 Marks]

- 1.5 marks → • Ideal DTFT amplitude need to be drawn
- 2 marks → • No, physically not realizable. Need to say/derive that the impulse response is leading to of length of infinite.
- 1.5 marks → • Methods are windowing, frequency sampling.

4. A continuous time signal sampled with 16000Hz is given to an ideal digital filter (whose impulse response is $h[n]$) to suppress the components of frequencies above 4000Hz. Compute the $h[n]$.

[5 Marks]

$$F_s = 16000 \text{ Hz}$$

$$F_c = 4000 \text{ Hz}$$

To suppress frequencies above 4000Hz, we need an Ideal Low-Pass Filter.

Discrete-time Cutoff Frequency: $\omega_c = \frac{2\pi F_c}{F_s} = \frac{2\pi(4000)}{16000} = \frac{\pi}{2}$ radians/sample ← 2 marks

Impulse Response: $h[n] = \frac{\sin(\omega_c n)}{\pi n} = \frac{\sin(\frac{\pi}{2}n)}{\pi n} = \frac{1}{2} \text{sinc}\left(\frac{n}{2}\right)$ ← 2 marks

5. Draw a flowchart to compute periodicity of a discrete-time cosine signal using correlation function?

[5 Marks]

3 marks → Compute Autocorrelation $R_{xx}[m]$ → Find first peak after $m=0$ → Peak location N is the period → End [Need to justify why is this]

↑ 2 marks

6. Compute the DTFT of $2\cos(\frac{\pi n}{3}) + 3\sin(\frac{\pi n}{6})$?

[5 Marks]

3 marks { Need to derive one of the following is the fine: $\cos(\omega_0 n) \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$ and $\sin(\omega_0 n) \leftrightarrow \frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

$$X(\omega) = 2\pi \left[\delta\left(\omega - \frac{\pi}{3}\right) + \delta\left(\omega + \frac{\pi}{3}\right) \right] + \frac{3\pi}{j} \left[\delta\left(\omega - \frac{\pi}{6}\right) - \delta\left(\omega + \frac{\pi}{6}\right) \right]$$
 ← 2 marks.

7. For the given system whose input and output relation is $y[n] = ax[n] + c$, then check whether the system is linear or not? Show the necessary computations.

[5 Marks]

A system is linear if it satisfies **Superposition** $T\{x_1 + x_2\} = T\{x_1\} + T\{x_2\}$ and

Homogeneity $T\{kx\} = kT\{x\}$

- Let $x_3[n] = x_1[n] + x_2[n]$

- $y_3[n] = a(x_1[n] + x_2[n]) + c = ax_1[n] + ax_2[n] + c$ ← 2 marks

- Sum of individual outputs: $y_1[n] + y_2[n] = (ax_1[n] + c) + (ax_2[n] + c) = ax_1[n] + ax_2[n] + 2c$

- Conclusion:** Since $y_3[n] \neq y_1[n] + y_2[n]$, the system is **Non-Linear**. ← 1 mark

8. Why is discrete-time frequency always rational and less than 1? Justify this with necessary derivations.

[5 Marks]

2.5 marks { **Rationality:** For a signal to be periodic in discrete time, the condition $x[n + N] = x[n]$ must hold. This requires $2\pi f N = 2\pi k$ (where k is an integer), hence $f = \frac{k}{N}$ must be a rational number.

2.5 marks { **Range (\$ < 1\$):** Because of sampling, to avoid aliasing it is within range of $[0, 1]$ or $[-0.5, 0.5]$

↑ more weightage
↓ may be (1.5 marks)

9. A system with the following input-output pairs has been observed during the operation of linear system. Can you draw any conclusions about the time invariance of this system? Justify your answer clearly.

[5 Marks]

$$\begin{aligned} x_1[n] &= \{-1, 2, 1\} \rightarrow y_1[n] = \{1, 2, -1, 0, 1\} \\ x_2[n] &= \{1, -1, -1\} \rightarrow y_2[n] = \{-1, 1, 0, 2\} \\ x_3[n] &= \{0, 1, 1\} \rightarrow y_3[n] = \{1, 2, 1\} \end{aligned}$$

[The solution is same as in Assignment3 Q2]

10. Let $x_1[n] = \cos(\frac{2\pi n}{N})$; $x_2[n] = \sin(\frac{2\pi n}{N})$; $0 \leq n \leq N-1$, then determine the N-point circular convolution and show the necessary computations?

[10 Marks]

[The solution is the same as in Assignment3 Q6]

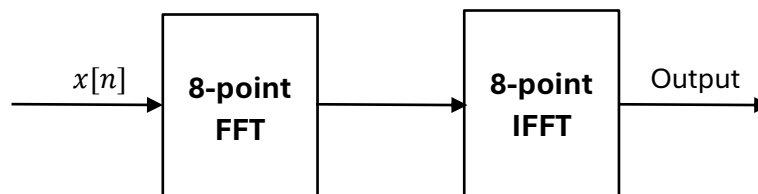
11. Let $x[n] = \{1, 2, 3, 2, 1, 0\}$ and $v[n] = \{3, 2, 1, 0, 1, 2\}$ then find the relation between DTFT of $x[n]$ and 6-point DFT of $v[n]$? Justify your answer with necessary calculations.

[5 Marks]

[The solution is the same as in Assignment2 Q7]

12. Let $x[n] = \{1, 2, 3, 4, 5, 6, 7, 8\}$, then compute output from the below block diagram. Justify your answer clearly.

[5 Marks]

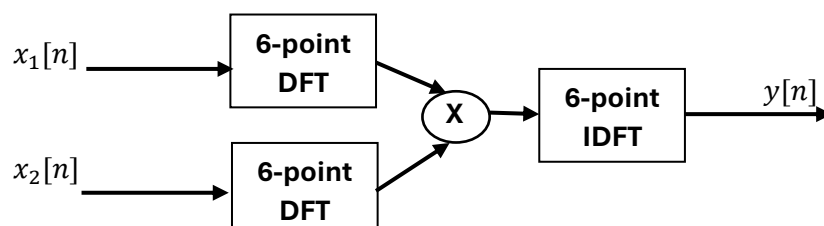


- Output is $x[n+2 \text{ modulo } 8]$, so the output is $\{3, 4, 5, 6, 7, 8, 1, 2\}$

- Need to justify the answers by saying:
 - FFT is equivalent to DFT
 - IDFT results modulo operation.

13. Let $x_1[n]$ and $x_2[n]$ are $\{1, 2, 1\}$ and $\{1, 2, 2, 1\}$ respectively then both $x_1[n]$ and $x_2[n]$ undergone process as shown in Figure below and result $y[n]$. Find $y[n]$?

[5 Marks]



Property: Multiplication in the DFT domain corresponds to **Circular Convolution** in the time domain. [Good to state this] $\leftarrow 1.5 \text{ marks}$

$x_1[n] = \{1, 2, 1, 0, 0, 0\}$ (Zero-padded to length 6)

$x_2[n] = \{1, 2, 2, 1, 0, 0\}$ (Zero-padded to length 6)

Perform 6-point circular convolution: $y[n] = \{1, 4, 7, 8, 5, 2\}$ $\rightarrow 2 \text{ marks}$

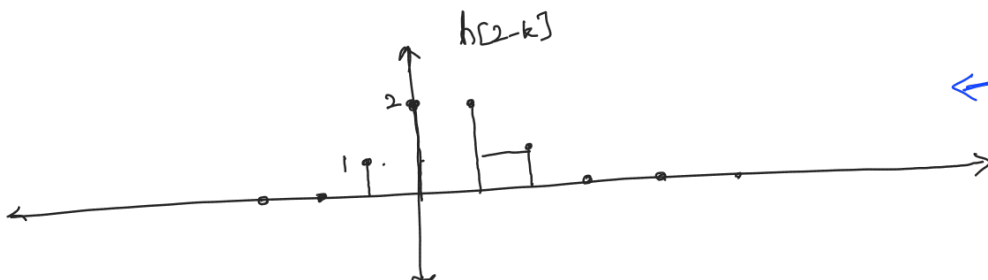
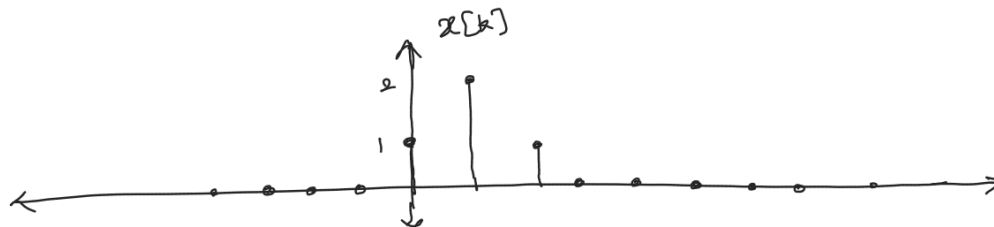
14. Let $h[n]$ and $x[n]$ are $\{1, 2, 1\}$ and $\{1, 2, 2, 1\}$, then $y[n] = x[n] * h[n]$. Write the equation involved in computing $y[2]$ and show its computation with the graphical illustration involving discrete-time signal representations?

[5 Marks]

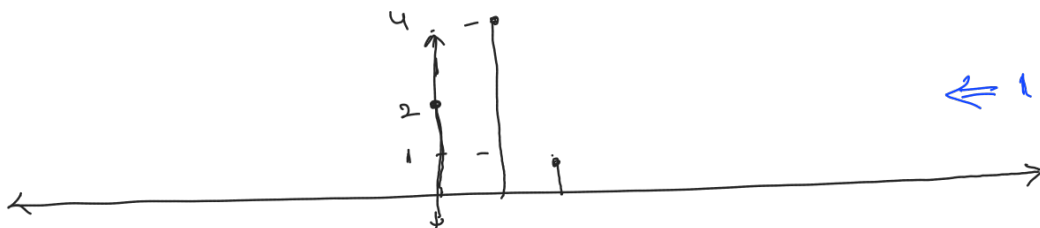
[Showing for the case of $y[2]$ is must, showing all the cases should result into significant reduction in the marks]

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[2-k]$$

$\leftarrow 1.5 \text{ marks}$



$\leftarrow 2 \text{ marks}$



$\leftarrow 1 \text{ marks}$

$$2 + 4 + 1 = 7 \quad \leftarrow 0.5 \text{ marks}$$

The reduction should be 3.5 marks: